

Graph Reductions

Compact Actual Proofs - Exam Ready

Proof Format

For each reduction, write only:

$$\boxed{\text{Build}} \rightarrow \boxed{\Rightarrow} \rightarrow \boxed{\Leftarrow} \rightarrow \boxed{\text{Time}} \rightarrow \boxed{\text{Result}}$$

The important line is always

$$\boxed{\text{source is YES} \iff \text{target is YES.}}$$

Reductions to Memorize

$$\text{VERTEX-COVER} \leftrightarrow_p \text{INDEPENDENT-SET}, \quad \text{INDEPENDENT-SET} \leftrightarrow_p \text{CLIQUE}$$

$$\text{VERTEX-COVER} \leq_p \text{SET-COVER}, \quad 3\text{-SAT} \leq_p \text{INDEPENDENT-SET}.$$

For the Set-Cover pair, the standard direct proof is $\text{VERTEX-COVER} \leq_p \text{SET-COVER}$.

1 Decision Problems at a Glance

Definitions

Problem	YES condition
VERTEX-COVER(G, k)	There is a set $C \subseteq V$ with $ C \leq k$ touching every edge.
INDEPENDENT-SET(G, k)	There is a set $S \subseteq V$ with $ S \geq k$ and no edge joins two vertices of S .
CLIQUE(G, k)	There is a set $Q \subseteq V$ with $ Q \geq k$ and every pair in Q is adjacent.
SET-COVER($U, \mathcal{S}, k$)	At most k sets from $\mathcal{S}$ cover all elements of U .
3-SAT(ϕ)	Some truth assignment satisfies every 3-literal clause of ϕ .

Two Lemmas That Do Most of the Work

$$\boxed{C \text{ is a vertex cover in } G \iff V - C \text{ is independent in } G}$$

$$\boxed{S \text{ is independent in } G \iff S \text{ is a clique in } \overline{G}}$$

2 Vertex Cover $\leftrightarrow$ Independent Set

Reduction

For $n = |V(G)|$:

$$\boxed{(G, k) \mapsto (G, n - k)}$$

The graph stays the same; only the parameter changes.

Actual Proof: VERTEX-COVER $\leq_p$ INDEPENDENT-SET**Build.** From VERTEX-COVER(G, k) output INDEPENDENT-SET($G, n - k$). $\Rightarrow$. Suppose C is a vertex cover with $|C| \leq k$. If two vertices outside C were adjacent, that edge would be uncovered. Hence $V - C$ is independent, and

$$|V - C| = n - |C| \geq n - k.$$

So the new INDEPENDENT-SET instance is YES.

 $\Leftarrow$. Suppose S is independent with $|S| \geq n - k$. No edge can have both endpoints in S , so every edge touches $V - S$. Thus $V - S$ is a vertex cover, and

$$|V - S| = n - |S| \leq k.$$

So the original VERTEX-COVER instance is YES.

Time. Only $n - k$ is computed, so the reduction is polynomial.**Result.**

$$(G, k) \in \text{VERTEX-COVER} \iff (G, n - k) \in \text{INDEPENDENT-SET}.$$

Reverse: INDEPENDENT-SET $\leq_p$ VERTEX-COVERUse the same transformation $(G, k) \mapsto (G, n - k)$.If S is independent with $|S| \geq k$, then $V - S$ is a vertex cover of size

$$|V - S| \leq n - k.$$

If C is a vertex cover with $|C| \leq n - k$, then $V - C$ is independent and

$$|V - C| \geq k.$$

Therefore

$$(G, k) \in \text{INDEPENDENT-SET} \iff (G, n - k) \in \text{VERTEX-COVER}.$$

Tiny ExampleFor the path $a - b - c$, $\{b\}$ is a vertex cover of size 1 and its complement $\{a, c\}$ is an independent set of size $3 - 1 = 2$.

3 Independent Set $\leftrightarrow$ Clique

ReductionConstruct the complement graph $\overline{G}$:

$$(u, v) \in E(\overline{G}) \iff u \neq v \text{ and } (u, v) \notin E(G).$$

Then use

$$(G, k) \mapsto (\overline{G}, k).$$

Actual Proof: INDEPENDENT-SET $\leq_p$ CLIQUE**Build.** From INDEPENDENT-SET(G, k) output CLIQUE($\overline{G}, k$). $\Rightarrow$. Let S be independent in G . No pair in S is an edge of G , so every pair becomes an edge of $\overline{G}$. Hence S is a clique in $\overline{G}$. $\Leftarrow$. Let S be a clique in $\overline{G}$. Every pair in S is an edge of $\overline{G}$, so none of those pairs is an edge of G . Hence S is independent in G .

Time. $\overline{G}$ is constructed by checking vertex pairs, in $O(|V|^2)$ time.

Result.

$$(G, k) \in \text{INDEPENDENT-SET} \iff (\overline{G}, k) \in \text{CLIQUE}.$$

Reverse: $\text{CLIQUE} \leq_p \text{INDEPENDENT-SET}$

Use the same map $(G, k) \mapsto (\overline{G}, k)$. A clique in G is exactly an independent set in $\overline{G}$. Therefore

$$(G, k) \in \text{CLIQUE} \iff (\overline{G}, k) \in \text{INDEPENDENT-SET}.$$

The construction is polynomial.

4 Vertex Cover $\leq_p$ Set Cover

Construction

Given $\text{VERTEX-COVER}(G = (V, E), k)$, create:

$$U = E$$

and for every vertex $v \in V$ create

$$S_v = \{e \in E : e \text{ is incident on } v\}.$$

Let $\mathcal{S} = \{S_v : v \in V\}$ and keep the same k .

Actual Proof

$\Rightarrow$. Suppose C is a vertex cover with $|C| \leq k$. Choose the sets S_v for all $v \in C$. Every edge has an endpoint in C , so every element of $U = E$ appears in at least one chosen set. Thus these at most k sets cover U .

$\Leftarrow$. Suppose at most k sets $S_{v_1}, \dots, S_{v_r}$ cover $U = E$. Let

$$C = \{v_1, \dots, v_r\}.$$

Every edge e belongs to some chosen S_{v_i} , which means e is incident on v_i . Hence every edge is touched by C , so C is a vertex cover and $|C| = r \leq k$.

Time. We create one set per vertex and list its incident edges; this is polynomial in $|V| + |E|$.

Result.

$$(G, k) \in \text{VERTEX-COVER} \iff (U, \mathcal{S}, k) \in \text{SET-COVER}.$$

Tiny Example

If $E = \{ab, bc\}$, then

$$U = \{ab, bc\}, \quad S_a = \{ab\}, \quad S_b = \{ab, bc\}, \quad S_c = \{bc\}.$$

Choosing vertex b is the same as choosing set S_b .

5 3-SAT $\leq_p$ Independent Set

Construction

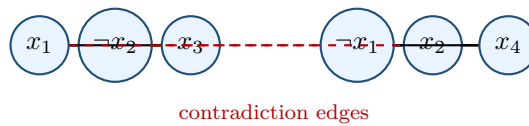
Let

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m.$$

Build graph G :

- one vertex for each literal occurrence;
- the 3 vertices of each clause form a triangle;
- connect complementary literals from different clauses;
- set $k = m$.

$$C_1 = (x_1 \vee \neg x_2 \vee x_3) \quad C_2 = (\neg x_1 \vee x_2 \vee x_4)$$



Actual Proof

$\Rightarrow$. Suppose ϕ is satisfiable. From each clause, choose one literal that is true. This gives exactly m vertices.

No two chosen vertices are adjacent:

- only one was chosen from each clause triangle;
- two true literals under one assignment cannot be complementary.

Therefore the chosen vertices form an independent set of size m .

$\Leftarrow$. Suppose G has an independent set S of size m . A clause triangle contributes at most one vertex. Since there are m triangles, S must contain exactly one vertex from each clause. Also, S cannot contain complementary literals because those vertices are connected.

Set every selected literal to true. The choices are consistent, and every clause contains one selected true literal. Hence ϕ is satisfiable.

Time. There are $3m$ vertices and at most polynomially many edges.

Result.

$$\phi \in \text{3-SAT} \iff (G, m) \in \text{INDEPENDENT-SET}.$$

6 Final Memory Sheet

One-Line Proof Triggers

Reduction		Remember this
VERTEX-COVER	$\leftrightarrow$	complement of a vertex cover is an independent set; use $n - k$.
INDEPENDENT-SET		
INDEPENDENT-SET	$\leftrightarrow$	complement the graph; keep k .
CLIQUE		
VERTEX-COVER	$\leq_p$	universe = edges; one set per vertex.
SET-COVER		
3-SAT	$\leq_p$	one triangle per clause; contradiction edges; target $k = m$.
INDEPENDENT-SET		